

Week C. By Deng Yufan.

Prob 1. Let G be a pseudo-random generator of $\ell(n) \geq 2n$.

* a. define $G'(s) = G(s0^{|s|})$. Is G' necessarily a PRG?

No. Let H be a PRG of $\ell'(n) \geq 2n$.

$$G(st) \quad (|s|=|t|=n) = \begin{cases} 0^{\ell(2n)} & \text{if } t=0^n \\ H(st) & \text{if } t \neq 0^n. \end{cases}$$

$$G(s) \quad (|s| \text{ is odd}) = H(s).$$

then $\ell(n) \geq 2n$. Obviously $G'(s) = 0^{\ell(2n)}$ is not a PRG.

Suppose G is not PRG, then $\exists A$.

$$\left| \Pr_{|s|=|t|=n} (A(G(st))=1) - \Pr_{|w|=p(2n)} (A(w)=1) \right| = \frac{1}{\text{poly}(n)}.$$

We have

$$\Pr_{|s|=|t|=n} (A(G(st))=1) = \frac{1}{2^n} \Pr_{|s|=|t|=n} (A(0^{\ell(2n)})=1) + \left(1 - \frac{1}{2^n}\right) \Pr_{\substack{|s|=|t|=n \\ t \neq 0^n}} (A(H(st))=1).$$

$\downarrow$ $O(2^{-n})$ $\downarrow$ $O(2^{-n})$

$$\text{Thus } \Pr_{|s|=|t|=n} (A(G(st))=1) = \Pr_{\substack{|s|=|t|=n \\ t \neq 0^n}} (A(H(st))=1) + O(2^{-n}).$$

$$\text{Likewise, } \Pr_{|s|=|t|=n} (A(H(st))=1) = \Pr_{\substack{|s|=|t|=n \\ t \neq 0^n}} (A(H(st))=1) + O(2^{-n}).$$

$$\text{So } \Pr_{|s|=|t|=n} (A(G(st))=1) = \Pr_{|s|=|t|=n} (A(H(st))=1) + O(2^{-n}).$$

$$\text{Thus } \left| \Pr_{|s|=|t|=n} (A(H(st))=1) - \Pr_{|w|=p(2n)} (A(w)=1) \right| = \frac{1}{\text{poly}(n)}.$$

Contradiction.



U b. define $G''(s) = G(s, s_2 \dots s_{n/2})$. Is G'' PRG?

Yes. We have

$$\begin{aligned} & \left| \Pr_{|s|=n} (A(G''(s)) = 1) - \Pr_{|w|=P(n/2)} (A(w) = 1) \right| \\ &= \left| \Pr_{|t|=n/2} (A(G(t)) = 1) - \Pr_{|w|=P(n/2)} (A(w) = 1) \right| \\ &\leq \text{negl}(n). \end{aligned}$$

Prob 2. A family of functions $\bar{F}_k$ is a pseudorandom permutation if

1. $\bar{F}_k$ and $\bar{F}_k^{-1}$ are efficiently computable.
2. $\left| \Pr_k (A^{\bar{F}_k, \bar{F}_k^{-1}}(1^n) = 1) - \Pr_{f_n} (A^{\bar{F}_k, \bar{F}_k^{-1}}(1^n) = 1) \right| \leq \text{negl}(n)$.

Consider enc. scheme:

1. Sample k uniformly.
2. On plaintext $x \in \{0,1\}^{n/2}$, Enc $_k$ samples $r \in \{0,1\}^{n/2}$, output $\bar{F}_k(r||x)$.

a. show how Dec $_k$ works.

For input y :

1. get $r||x = \bar{F}_k^{-1}(y)$.
2. output x .

b. Prove the enc. scheme is CPA-secure.

Suppose we use random f_n instead of $\bar{F}_k$, then $\Pr(A \text{ succ}) \leq \frac{1}{2} + \frac{\text{poly}(n)}{2^n}$.

Suppose $\exists A$, $\Pr(A \text{ succ}) \geq \frac{1}{2} + \frac{1}{\text{poly}(n)}$, then $|\Pr(A \text{ succ}) - \Pr(A \text{ succ})| \geq \frac{1}{\text{poly}(n)}$.

Construct Distinguisher D for $\bar{F}_k$ using A : Suppose A has Enc $_k(w)$ as Oracle.

D give result as Enc $_k(w) = \bar{F}_k(r||w)$, where r is random.

if A right, output 1, otherwise 0.

Obviously $\Pr(D^{\bar{F}_k, \bar{F}_k^{-1}}(1^n) = 1) = \Pr(A \text{ succ})$, $\Pr(D^{f_n, f_n^{-1}}(1^n) = 1) = \Pr(A \text{ succ})$.

So D can distinguish $\bar{F}_k$ and f_n , contradiction.

